Roll No. | | | | | | | |

End Semester Examination 2025

Name of the Course: B. Tech (EE) Semester: IV
Name of the Paper – Introduction to Electrical Energy Sources
Paper Code: TEE 402

Time: 3Hours Maximum Marks: 100

Note: -

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q. 1	(20 Marks)	
(a)	Explain the working of economizer, superheater, and air preheater in a thermal power plant. How do they improve efficiency?	CO2, CO3,
	OR	
(b)	What are the different types of boilers used in thermal power plants? Explain any one with a neat sketch.	CO2, CO3,
	OR	
(c)	Explain the function and working of a condenser in a thermal power plant. Why is it essential?	CO2, CO3,
Q. 2	(20 Marks)	
(a)	Explain (a) Tailrace (b) Penstock (c) spillway	CO2, CO3, CO4
	OR	
(b)	What are the different types of hydro turbines? Classify them based on their different characteristics. Differentiate between the operation of a Pelton turbine and a Francis turbine.	CO2, CO3, CO4
	OR	
(c)	The total water inflows for different months for a river are as under. Jan: 1000×10^6 m^3; Feb: 800×10^6 m^3; March: 550×10^6 m^3; April: 300×10^6 m^3; May: 300×10^6 m^3; June: 1500×10^6 m^3; July: 3000×10^6 m^3; Aug: 3000×10^6 m^3; Sep: 2500×10^6 m^3; Oct: 1500×10^6 m^3; Nov: 1450×10^6 m^3; Dec: 1600×10^6 m^3 (a) Draw a hydrograph from this data and find out the average monthly flow. Assuming that all the inflow is used for power generation, how much power can be generated if available head is 100m and overall plant efficiency is 80%.	CO2, CO3, CO4
Q. 3	(20 Marks)	
(a)	Explain a fission nuclear reaction. Differentiate between uncontrolled and controlled chain reaction.	CO2, CO3, CO4 CO5
	OR	

(b)	(a) How mass and energy are related (b) When a U_{92}^{235} nucleus fissions, about 200 MeV of energy is released. What fraction of the weight of the nucleons is thereby converted into energy?	CO2, CO3, CO4 CO5
	OR	
(c)	Explain the terms (a) Radioactivity (b) Multiplication factor (c) Moderator (d) Fertile and fissile materials	CO2, CO3, CO4 CO5
Q. 4	(20 Marks)	
(a)	Draw the layout of a medium size (a) Diesel electric plant (b) Gas turbine plant	CO2, CO3, CO4 CO5
	OR	
(b)	Draw a diagram showing an open cycle gas turbine plant with regeneration, intercooling and reheating.	CO2, CO3, CO4 CO5
	OR	
(c)	Name and briefly discuss the function of all the essential equipment's in a diesel electric plant.	CO2, CO3, CO4 CO5
Q. 5	(20 Marks)	
(a)	The load on a power plant on a typical day is as under Time: 12-5 am, 5-9 am, 9-6 pm, 6pm -10 pm, 10pm-12am Load (MW): 20, 40, 80, 100, 20 Plot the chronological load curve and load duration curve. Find the load factor of the plant and the energy supplied by the plant in 24 hours.	CO1, CO6
	OR	
(b)	The annual load duration curve of a small Hydro plant shows 438 $*10^4$ kWh of energy during the year. It is a peak load plant with 20% annual load factor. Find station capacity. If plant capacity factor is 15%, find reserve capacity of the plant.	CO1, CO6
	OR	
(c)	Explain the followings i. Load factor ii. Demand factor iii. Base load iv. Peak load v. Capacity factor.	CO1, CO6